RESEARCH AND IMPLEMENTATION CONTRACT

Latent Causal Infeasibility for Long-Horizon TAMP

Implementation and experimental guide v1.8

2D, 3D and migration to ordered repair

Implement Methodology v1.6. Preserve the completed geometry, robot and evidence work. Add explicit ordered proposals, intervention-effect outputs and motion-cost supervision before claiming that the learned framework produces explanations and executable corrective actions. Work one gate at a time.

Current reported position

G4A report 0008 records NOT_READY_TO_SCALE: three zero-move roots against a target of six. The geometric candidates are not learned. No G4B, model training, 3D or hardware result is established by that report. This revision does not claim that the repository was inspected or that any new experiment was executed.

The previously prepared G4A follow-up remains a narrow V1 evidence diagnostic. Finish or report it under its own frozen plan; do not enlarge it to collect V2 sequence data. The next new architecture checkpoint is G4D, followed by G4B sampler validation, G4C data readiness and G5 learning. G4D is an added migration checkpoint; the letter does not imply that it runs after G4C.

Core architectural change

The learned proposal is now an ordered sequence of object/destination pairs. Its score includes initial robot configuration and an order-sensitive recurrent model. The external backend may choose grasps and trajectories *within* that order, but cannot silently choose a different order. Explanations explicitly compare specified interventions; motion cost has a measured training target.

Document authority

Specification v1.6 defines the variables, equations, evidence and claims. This guide defines records, modules, defaults, gates and acceptance. Together they supersede v1.5/v1.7 for new primary learned work. Keep older protocol implementations, reports and run directories byte-unchanged where archived. New files and outputs require V2 metadata. The old master is motivation, not a conflicting implementation authority.

No bulk generation, training or live actuation is authorized merely by reading this guide. Each gate requires its stated acceptance evidence and a stage-specific run plan. Do not ask repeatedly for permission for work already authorized in that run plan.

1. Read the stages as questions, not as model layers

G0: Can we represent and reload the scene? Check identities, coordinates, supports, robot state and visual agreement. Output the full scene and its records.

G1: Can the complete robot perform one simple skill? Validate FK/IK, links/gripper, attachment, continuous paths, release and retreat. The checked motion, not a moving object icon, is the evidence.

G2: Can we measure a controlled change? Historical fixed-motion interventions checked local obstruction and geometry. Retain that scope; a blocked reference path does not prove task impossibility.

G3: Can valid endpoints actually be reached? Historical assignment and order verification distinguished terminal geometry from executable sequences. Preserve its analytic baselines and scope.

G3.5: Does the trajectory-free one-shot interface work? Manual candidates, achieved-state target planning, no target feedback and explicit unknowns. Historical V1 ordering was external.

G4A and its bounded follow-up: Is automatic evidence trustworthy and affordable? Reconcile certificates, calls, coverage and costs. Keep the original gate failure visible.

G4D: Can we express and supervise the newly required outputs? Add ordered records, fixed-order verification, counterfactual explanations and cost provenance on sparse cases. This must precede V2 learning.

G4B: Is the joint sampler mathematically correct? Tiny exhaustive sequence spaces test destinations, ordering and variable-length moves.

G4C: Is the V2 dataset ready? Audit ordered labels, explanation pairs, cost targets, splits and throughput before a 300-root pilot.

G5: Can the networks learn useful scores? Train static, task, effect and cost objectives, then joint feasibility ranking. Inspect scene-backed validation behavior.

G6: Which components help? Matched baselines, order/robot/effect/cost ablations, held-out roots and all-episode denominators.

G7–G9: Does the same contract work with the full 3D robot? Validate sparse cupboard skills, complete ordered repairs, then separate 3D data/training/evaluation.

G10: Do local repairs help a longer task? Carry achieved states through a scripted sequence and measure downstream interference. Hardware remains a separate authorized evidence level.

2. Minimal implementation ownership

Inspect actual repository instructions and files first. Do not rebuild working modules to match an ideal tree. Prefer ordinary modules around 120–280 logical lines; roughly 300 is a review trigger, not a reason to split functions arbitrarily. No generic framework, database, nested package hierarchy or separate 3D training stack.

Scene/configuration modules own immutable initial state, structured action/goal, candidates, robot profile and strict schema. **Existing geometry, motion and skill modules** retain FK/IK, collision checks and continuous skill execution.

Existing proposals/solve modules own canonical ordered candidates, physical signatures, costs derived from endpoints, ranking and the V2 sampler. Preserve frozen A0 and V1 behavior behind explicit protocol routing. Split sampler code into one focused module only if the current file would otherwise become unclear.

verify.py adds fixed-order verification without removing historical order search. **evaluate.py** owns commitment, execution and post-repair target evaluation. It never lets the proposal module import the target callback.

interventions.py or its existing equivalent creates valid fixed-robot pose substitutions and changed-field audits. **data.py** constructs allowlisted tensors, canonical prefix geometry, masks, scope-aware labels and root splits.

model.py owns frozen feature pooling, static heads, scene/context encoding, ordered GRU, task and cost readouts. **learn.py** owns losses, optimization and checkpoint compatibility. Create these only at G5 except for a minimal deterministic scoring interface required by G4B.

Existing audit/certificate/coverage modules own independent checking, negative scopes, readiness and budgets. **Existing visualization modules** draw saved states, predicted ghosts, order, explanations and replay. Do not add a web dashboard. **run.py** remains a thin dispatcher.

Small interfaces, explicit boundaries

Use equivalents of `score_sequence(context,sequence)`, `explain(context,sequence,budget)`, `plan_fixed_order(state,sequence,budget)`, and `evaluate_target(achieved_state,action,budget)`. The first two cannot access oracle calls. Fixed-order planning has no target callback and returns a full witness or an unresolved/certified result with scope.

Keep JSON/NPZ records and flat per-run outputs. A source file's name or line count is not a scientific contribution; optimize clarity and auditability.

3. Records and migration without label corruption

Query and learner context

Store protocol, schema version, root/query metadata, immutable initial scene, action/goal, constraints, physical candidate maps, initial robot/gripper state, mask/pose provenance and profile hash. Root IDs are split metadata only. The learner consumes a strict allowlist, not a flattened record.

Ordered proposal

Store `sequence=[(object_id, physical_pose_id), ...]`, derived assignment, moved count, initial-context hash, source, proposal seed, task score, feasibility/sample energy, known endpoint distance, predicted joint cost and frozen rank. Reject repeated IDs, stay entries in the action list, unknown poses and length above the cap. Empty list is the unique no-repair proposal.

Evidence

Separate `static_status`, `ordered_feasibility_status`, `relocation_attempt_status`, `target_attempt_status`, `simulation_status` and `hardware_status`. Each certificate identifies property, domain, sequence/assignment scope, initial state/action/profile hashes and independent checker result. Trajectory failure is not automatically an ordered negative.

Cost and explanation

Cost records contain policy ID, first relocation-witness hash, full start/end joints, unwrapped path, joint ranges, measured normalized travel and label availability. A successful relocation with an unresolved target may still have a valid relocation-cost label.

Explanation records contain kind (scene substitution or repair omission), object set, changed poses/actions, baseline and alternative contexts, fixed robot-state hash where applicable, predicted scores/effect, eligibility, label mask and independent evidence references. Preserve the original appearance-feature anchor for both sides of scene comparisons.

Commitment and achieved execution

Freeze sequence, destinations, chosen relocation witness, rank and commit-event index. After each move store commanded and achieved object poses, joints, gap, attachment and cumulative moved IDs. Validate actual states before the next committed action. No resetting admitted deviations to commanded values.

Legacy reuse

Use a new migration manifest referencing original files by hash. A V1 complete witness can supervise its exact recorded ordered sequence only after policy/state/schema validation. Assignment-wide negatives transfer only within the same scope. Do not copy positive endpoint labels to every order. Do not infer the new first-witness cost target from a conveniently selected successful trace. Never overwrite old reports or regenerate their archives.

4. Configuration: proposed starts, not measured optima

Preserve the actual tested robot profile, 48-grasp family, home, gripper and tolerances. Do not copy provisional old defaults. Validate strict TOML keys and save resolved configuration, runtime and source hashes. These are initial V2 settings to test, not accepted performance settings.

```
[protocol]
name = "INITIAL_SCENE_ORDERED_ONE_SHOT_V2"
proposal_observation = "INITIAL_ONLY"
order_source = "PROPOSED_SEQUENCE"
executor_may_reorder = false
target_feedback_to_selection = false
max_primary_target_evaluations = 1
target_end_policy = "GOAL_AND_SAFE_RETREAT"
home_return = "SEPARATE_DIAGNOSTIC"
move_once = true

[model]
width = 128
sequence_encoder = "GRU"
visual_backbone_frozen = true
counterfactual_appearance = "INITIAL_ANCHOR"

[sampler]
move_cap = 3                # sparse V2 development
chains = 8
updates_per_chain = 300
temperature = 1.0
kernel_probabilities = [0.25, 0.25, 0.25, 0.25]
hard_static_mask_inside_chain = false
max_unique_candidates = 64
max_candidates_to_verify = 12

[energy]
lambda_task = 1.0
lambda_endpoint_distance = 0.1
lambda_predicted_joint_travel = 0.1
joint_training_energy = "FEASIBILITY_ONLY"

[selection]
task_threshold_source = "VALIDATION_FROZEN"
rank = ["move_count", "predicted_joint_travel",
        "endpoint_distance", "feasibility_energy", "signature"]
```

Pin grasp/motion/target node and evaluation limits from the actual tested backend in the stage plan. Do not interpret an absent threshold as zero; it blocks learned evaluation until chosen on validation. Separate sampler, model, root, relocation and target seeds; the latter two are not learner inputs. Larger move caps and different update counts are separately planned experiments.

5. Concrete tensors and the ordered network

Per-object data

Cache frozen mask-pooled visual features once per initial image. The trainable object projection has width 128. Append normalized initial pose, shape/support descriptors and numerical context; build destination-conditioned tokens for physical candidates. Missing features use a mask, not accidental zero-as-observed semantics.

Static unary and pair MLPs start with two hidden layers of width 128 and ReLU. Pair logits are exchange-symmetric, for example by averaging the network on swapped arguments. Static heads do not consume teacher distances or trajectories. All present objects contribute, even those never moved.

Prefix builder

Given a canonical sequence, build ideal scene geometry before each action by applying preceding endpoint substitutions to a clone of initial poses. The currently selected object's origin is its initial pose because it has not moved before. Its destination may be initially occupied by an object that has already moved. Fixed workspace and the resting target remain present.

Pool all prefix object tokens with a shared MLP and sum/count aggregation. Concatenate the pooled prefix, selected object/destination token, action index/length and initial robot/action context. Feed these tokens to a single GRU of width 128. Initialize its hidden state from the initial scene/context and read out using the final hidden state, final scene pool and length. Padding is masked and does not become a decision.

Readouts and empty sequence

A linear/MLP task readout returns a logit. A cost readout uses softplus for nonnegative normalized joint travel. Empty repair uses the initial/final scene with no recurrent updates and has cost exactly zero. No separate feasibility classifier is required: the empty sequence score is the initial-task prediction.

Counterfactual explanation

Keep original object appearance vectors fixed. Replace structured poses in a valid diagnostic clone, rebuild context/prefix geometry and score the empty sequence. For omission, remove the action and rebuild every affected prefix from the original initial state. Do not delete the object slot, zero its visual feature, or retain its proposed endpoint after omitting its move.

The GRU hidden state is not a predicted joint state. Teacher intermediate joints are evidence and cost supervision only. The model predicts prospects from initial joints and the proposed sequence; actual intermediate joints come from continuous planning. If future work learns joint transitions, it requires a separately validated model and must not be assumed here.

6. Sampler algorithm and exact correctness tests

Implement the methodology's distribution on canonical sequences. A toy score is sufficient at G4B; no neural training or target planner is needed. Store physical pose signatures; never use object-name sorting as a causal heuristic.

```
for chain in seeded_initial_sequences:
    r = chain.initial_sequence
    for step in range(max_updates):
        kernel = choose_fixed_probability_kernel()
        if kernel == "destination":
            choose an action position uniformly
            score every nonstay destination with FULL energy
            sample categorical softmax(-energy / temperature)
        elif kernel == "swap":
            choose two action positions uniformly
            propose swapped sequence
            accept with symmetric Metropolis probability
        elif kernel == "birth":
            choose unused object, destination, insertion position
            accept with full Metropolis-Hastings ratio
        else:  # death
            choose action position and remove it
            accept with reverse/forward proposal ratio
        unavailable kernel => self-transition
        collect r according to fixed schedule
deduplicate ordered physical signatures
static-screen, threshold and freeze ranking
```

Use log probabilities to avoid overflow. Do not renormalize kernel probabilities at boundaries unless the Hastings calculation includes that change. Birth/death ratios in the specification assume fixed equal kernel probabilities and uniform choices. Variable candidate counts K_i must be included.

Required exact test

For two or three objects, two destinations each and move cap two, enumerate every canonical sequence, including empty. Compute exact normalized Boltzmann probabilities. Build the transition matrix from the actual kernel probabilities and check row sums, detailed balance and stationary distribution to numerical tolerance. Use asymmetric destination counts in a second case. A fixed-seed histogram is supplementary, not the only mathematical test.

Test full-energy conditional differences, correct ordered deduplication, birth/death reversibility, one-object/no-candidate/empty boundaries, slot relabelling and fixed-seed reproducibility. Check optimized prefix caches against full recomputation. Instrument that scoring cannot call a planner. Temporary static conflicts may occur in chains; no invalid candidate passes the final external screen.

7. Teacher labels, matched interventions and cost records

The offline teacher is a distinct orchestration path allowed to call relocation and target planning. Its records must never be flattened into learner tensors. A query pool includes several ordered proposals, preferably multiple successful alternatives and certified negatives beyond trivial endpoint overlap.

For each r, restart from the immutable initial state. Ask the fixed-order backend to find a complete relocation witness. Store its first complete witness and normalized joint travel before examining target success. Evaluate the target from its achieved final state. Additional offline realizations may establish existence, but do not replace the first-witness cost target with a target-selected path. Record schedules and all attempts.

Positive ordered labels require a complete repair-plus-target witness. Negative labels require a scope-wide proof for the order; a first-action pose-only exclusion can qualify if every permitted grasp/placement realization is excluded. Search exhaustion, failed connectors from one chosen grasp and numerical ambiguity remain unknown.

For diagnostic scene pairs, substitute poses while keeping robot joints, gap, attachment, task and other fields fixed. Check that both states are well-defined. Ineligible counterfactuals are reported separately. Keep initial visual features anchored for the learner on both sides; rendering the counterfactual is for visualization only. Label target-only feasibility independently. Train the effect difference only if both outcomes are decided.

For omission pairs, remove an action, restore its object to stay, preserve the remaining action order and regenerate the whole sequence evidence. Do not reuse the old remaining trajectories without checking them in the changed scene. Joint omission tests are needed for interacting objects.

Keep the evidence masks separate

An unknown ordered outcome can still have decided static labels and a complete relocation-cost label. An unknown baseline target prevents a definite causal-effect label even if the intervened scene succeeds. A positive original order does not make its omissions negative. A certificate may carry several reasons; count unique assignments/sequences separately from accepted certificate records.

Stop condition

If the teacher cannot produce enough decided order contrasts or matched effect pairs, do not train and call the result order-aware causal repair. Report the missing coverage, construction attempts and actual CPU cost. Do not make the planner more permissive or relabel timeouts to meet a quota.

8. Preserve G0–G3.5 and complete the existing G4A follow-up

Read reports and instructions in the repository before touching code. Record branch, commit, working tree, current profile/configuration and archive hashes. The reported G4A commit is `44679ce67f2ffdf88afdcf3c58e09b48c6dbd76f`; verify actual state rather than resetting to it. Preserve unrelated local edits and never force-push.

Historical facts are not new measurements

Report 0008 states 20 roots, 240 assignments, 44 positives, 86 negatives, 110 unknowns; three zero-move roots; 8/20 geometric primary success; and 4.56 CPU-hours against a 4.0 cap. It reports 94 tests and kinematic replay, not hardware robustness. Preserve these records and their original protocol.

The new reporting reconciliation checks 89 certificate records versus 86 unique negative assignments, exact overlaps/duplicates, and at-most-one target-call accounting. Episodes without repair or with abort must not be described as having one target call. Unknowns split into 67 relocation-search exhaustion, 29 target-budget exhaustion and 14 other assignment-level unknowns; most are not target-budget misses.

Bounded diagnostic already specified

The separate follow-up instruction tests all-stay on frozen r04, r14 and r15, with B1/B2 target-only work allowances, at most six new evaluations, 900 CPU-seconds per evaluation and 7,200 aggregate planning CPU-seconds. Keep that instruction's state, seed, cap, scheduling and stopping rules. Do not re-run all 240 assignments or replace the roots based on outcomes.

Track CPU during work, not only at root completion. Record in-flight work, cancellations and censoring. A dropped-root sensitivity calculation cannot reconstruct a cap-compliant historical run without scheduling logs. Seed-panel results on nine executed cases do not establish that seeds never help.

This follow-up stays V1 and retains the original NOT_READY_TO_SCALE decision. Even three new zero-move witnesses would be diagnostic-enriched coverage, not evidence that V2 sequence/explanation training is ready. Its results may inform the next frozen data plan.

Compatibility tests

Keep old replay adapters scoped to the recorded policy and log any filled config keys. New config sections must not alter old semantics. Compare archive hashes before/after, and rerun required historical replay gates in separate output paths. No broad refactor is justified by this architecture revision.

9. G4D: new output and ordered-evidence migration gate

Purpose

Make every new requirement concrete before writing a trained model. This gate uses manually specified or deterministic ordered candidates. It validates representation, evidence and execution boundaries, not learned performance. Run it after reporting the bounded G4A follow-up; it does not erase that gate's original decision.

Implementation steps

Add V2 protocol/schema, canonical sequence validation and derived assignments. Extend the verifier with a fixed-order mode and reject silent reordering. Add anchored-appearance counterfactual inputs, scene/omission evidence records, normalized path-cost calculation and sequence/explanation visualizations. Keep V1 order search callable only in V1 or explicitly named baseline paths.

Required sparse cases

D-V2-0: a checked empty-repair target success, with exact zero relocation cost. **D-V2-1:** a mandatory goal occupant and a checked sufficient repair, with original/edited target-only evidence where both are valid.

D-V2-2: identical endpoints under two orders: one complete success and one order-wide certified negative. A timeout in the bad order is retained but does not satisfy the decided-order contrast. If construction fails, mark this requirement BLOCKED.

D-V2-3: destination choice changes whether another object must move; retain alternatives and occupancy constraints. **D-V2-4:** a joint intervention/omission comparison exposing an interaction that singleton scores alone do not summarize.

D-V2-5: complete relocation traces with computed joint travel and at least two alternative proposals for comparing endpoint distance with measured robot cost; no predetermined inequality is assumed. **D-V2-6:** a robot-start-state counterfactual with identical object geometry and at least one independently decided outcome per valid start. Different outcomes are desirable evidence, not fabricated labels.

Acceptance and stopping

Every case records attempted, decided, unknown or unavailable status with complete witnesses or certificates. At least one checked nonempty two-move sequence, one decided same-endpoint order contrast, one decided scene-effect pair and one decided omission pair are blocking requirements. Real intermediate joint continuity and target-start equality must be demonstrated. Synthetic scores cannot satisfy these robot-evidence requirements.

Freeze case search limits and a total CPU cap before collection. Keep all attempted constructions. No undocumented hardware run, new action type or relaxed collision rule to make a case pass. Produce a migration report and stop before G4B.

10. G4B and G4C: correctness before scale

G4B sampler gate

Use the exact tests in Section 6. Demonstrate an order-dependent synthetic score, a destination interaction, and different candidate counts. Show energy histories and visited canonical sequences, marking the display as a mathematical test rather than robot evidence. Sampler correctness may be studied on a finite toy domain, but no downstream claim passes without G4D's physical evidence.

G4C ordered data audit

Create a new development-only 20-root plan, rather than silently reusing G4A as a V2 benchmark. Predeclare the strata and attempt ledger. A starting construction mix is six intended zero-repair controls, four goal-occupancy cases, four access/order cases, four competing-destination cases and two joint-interaction cases. These are construction intents, not truth labels; failures remain in the ledger.

For each root, propose up to 12 endpoint assignments and up to three canonical orders per nonempty assignment, including empty once. Freeze candidates before target outcomes. Record both assignment and sequence counts. Select diagnostic singleton/joint edits and omissions by a frozen geometry/sequence rule, not by hindsight success.

Proposed predeclared readiness targets

Require 20 audited roots, at least six roots with zero-move witnesses, eight with nonempty complete witnesses and four with multiple sufficient proposals. Require at least four roots with decided same-endpoint order contrasts, four with decided nonzero scene effects and four with decided omission effects. Include at least four decided zero-effect/no-change or nuisance control roots. Require motion-cost labels on at least eight roots, including multi-move traces on four.

Require informative static-valid certified task negatives on at least three roots beyond pure goal occupancy, no semantic/protocol contradictions and at most 50 percent unknown among static-valid ordered candidates. Report eligibility and unknown coverage separately for effect pairs; require at least 50 percent decided among eligible planned pairs. These are proposed development thresholds, not a proof of statistical adequacy or an assertion they are attainable.

Set actual teacher schedule, maximum construction attempts and CPU cap in the stage plan. They cannot be sensibly inferred from the old 4-hour run for this larger label workload. Report BLOCKED when targets cannot be met; revise only a future plan, preserving the failed cohort.

Only after this audit passes, split a 300-root pilot before descendants (210/45/45), then freeze test identities. Conditional 1,000-root expansion requires measured throughput and useful evidence. Do not use development roots as untouched test roots.

11. G5: training sequence and focused checks

Use the architecture and losses in Specification v1.6. Proposed starts: frozen DINOv2 ViT-S/14 or documented equivalent; width 128; AdamW learning rate 10^{-3}, weight decay 10^{-4}; all loss coefficients 1; training pool temperature 1; mixed pools near 32 decided sequences when available; three independent model seeds. Pin actual device/library compatibility after a smoke test, not an old unrelated CUDA configuration.

Training progression

First verify tiny-batch learning on clean decided labels without scientific claims. Train static, task, effect and cost objectives, monitoring them separately. Then add the finite-pool feasibility-energy loss on mixed positive/negative pools from the same initial query. The cost-augmented sampling energy is not the joint feasibility-loss energy. Save this distinction in the checkpoint schema.

Choose validation operating thresholds for initial infeasibility, repair eligibility and abstention. The repair threshold must prevent all-stay automatically winning on move count when predicted infeasible. Freeze hyperparameters and ranking before test. Do not choose them to reproduce the teacher's arbitrary preferred set.

Required checks

Validate BCE and joint-loss masks, finite-difference energy gradients, bounded effect differences, nonnegative cost, empty cost zero, and no effect loss when either side is unknown. Check scene substitution preserves visual anchor and robot state. Check omission restores the object's initial pose and regenerates later prefixes.

Object-order permutations must leave scores invariant under consistent remapping; action-order changes must remain visible to the network. Verify padding masks, variable object counts and candidate maps. A freshly initialized network need not assign a desired order preference; the test must verify representational sensitivity and then supervised behavior on a controlled case.

Poison oracle fields, root IDs, teacher states and reference paths and require identical learner tensors and predictions. Delete reference fields entirely. Check score/energy equality after checkpoint reload. Reject a checkpoint with mismatched protocol, robot/profile, feature preprocessing, schema or target-end policy.

Evidence to inspect

Show initial scenes, predicted no-repair scores, candidate ghosts and order arrows, scene/omission effects, teacher evidence, measured versus predicted joint travel, and complete held-out validation replays. Include reliability/coverage plots and category-wise errors, not just loss curves. If learned order or explanation does not beat its control, report that result; passing software tests alone does not establish the contribution.

12. G6: matched comparisons for every added claim

Keep historical B0–B9/O1/F0 IDs with their original definitions. Use descriptive V2 names in a separate result namespace: `V2_FULL`, `V2_GEOMETRY_ONLY`, `V2_UNIFORM_ORDERED`, `V2_ENDPOINT_EXTERNAL_ORDER`, `V2_FIXED_ORDER`, `V2_NO_EFFECT_LOSS`, `V2_NO_COST`, `V2_NO_JOINT_LOSS`. Record exact inputs, learned heads, sampling and executor policy for every method.

V2_ENDPOINT_EXTERNAL_ORDER is the previous architecture under its declared external order budget; its extra order search is counted. A fixed/random-order ablation tests ordered decisions, while a retrained order-insensitive head tests representation. Equal candidate counts alone do not make methods equally expensive: also report full network calls, motion attempts, target calls and CPU/time.

Output-specific metrics

For initial infeasibility, use only decided target-only labels for accuracy/calibration, and show unknown and abstention coverage. For explanations, measure scene-effect and omission-effect errors separately, signed effects, joint interactions and nuisance controls. If the model merely identifies moved participants, do not call that direct-cause accuracy.

For repairs, overall success denominator contains every eligible root/query, including abstention, no witness, abort and target unresolved. Also report relocation completion, conditional target success, moves, endpoint distance, measured joint travel and latency. Empty relocation counts as completed; zero conditional denominators are N/A.

For necessity/minimality, evaluate only on independently certified scopes. A complete witness is an upper bound on move count; a failed search is not a lower bound. For order ranking, use same-endpoint decided alternatives and report how often such contrasts were available.

Match information and computation

Use the same candidate domain, robot and target evaluator across primary methods. Freeze thresholds, caps and seeds before test. All target feedback occurs only after commitment and completed repair. O1 may use privileged target search offline, but does not compete as an equal-information learned baseline.

Root-cluster confidence intervals, failure-source breakdowns and costs are required. Match intervention/observational datasets on root families, labels and decided coverage before attributing a benefit to interventions. Compare pair inference lesions with retrained models. Hold data fixed for the joint-loss ablation. Keep robot geometry and joints in the full model; a robot-state-removal model is a diagnostic ablation, not an acceptable default.

13. G7–G9: the same decisions in the full 3D scene

G7: assets and basic actions

Pin full xArm7, gripper, joint/TCP conventions, mounting pose, valid home and collision/contact exclusions. Reject UNSET/UNPINNED keys. Keep supported upright transport: no lifting over obstacles. Verify bottom/support contact, tilt, yaw policy, attachment and retreat throughout trajectories. Arm links are free to move in 3D.

The proposed pilot retains a 0.64 m square cupboard, 0.005 m support top, 0.30 m roof underside and 0.010 m wall thickness; cylinders have radius 0.024 m and height 0.075 m. Their supported center is 0.0425 m. Validate these numbers with actual assets before freezing them. Proposed height/tilt tolerances are 0.001 m/0.035 rad, not guaranteed hardware tolerances.

Show top, front and side views of the full arm/gripper and the object height/tilt trace. Distinguish kinematic replay from dynamics and physical grasp stability. No planar surrogate can certify a 3D result.

G8: ordered repairs and achieved-state continuity

Implement sparse counterparts of G4D before density. Keep the proposed order fixed while searching grasps and paths. Demonstrate an alternative-order case, occupied-destination dependencies and actual final-state target planning. Check the next action from accepted achieved poses and joints; abort when assumptions fail. No connector to a historical frozen target start is used in V2.

Use 6–10 obstacles for the initial pilot, not an arbitrary five-blocker target. If a desired case cannot be constructed, retain that result. A finite IK failure remains unknown unless a complete scoped exclusion proof is available.

G9: separate training and matched evaluation

Reuse the same records, model and training modules with a suite configuration. Audit 20 3D roots and order/effect/cost coverage before a separate 300-root pilot. Train 3D independently first; weight transfer from 2D is an extra experiment. Recalibrate costs, thresholds and normalized geometry using training/validation data only.

Repeat the relevant V2 controls with the same target-feedback boundary. No certificate, cost scaling or physical robustness claim transfers automatically from 2D. Preserve simulator/asset hashes, contact conventions and assistance flags. Unknown shape, real mask/pose errors and physical slip remain separate limitations until tested.

14. G10, execution deviations and hardware scope

Long-horizon script

Use at least three original target actions with real achieved scene continuity. Obtain a fresh initial observation for each new target episode. Within one episode, freeze its ordered correction, selected objects and destinations. Reset move-once history only after successful completion or explicit abandonment, never to hide a second relocation of the same object.

Include a case where a locally useful correction obstructs a later task. The model has no future-plan-suffix input in this version. Report local action success separately from whole-script success, all repair invocations and extra observations. A target-feedback retry system is a separately named protocol.

Achieved-state policy

After each relocation, validate identity, robot configuration, gripper gap, attachment and object poses against the committed assumptions. Preserve admitted deviations in the scene. Re-certify or replan the same next action if the declared low-level policy permits; never silently restore nominal poses or reorder the high-level sequence. Record low-level attempts and their costs.

The reported G4A 1 mm/6 mm perturbations are fault-injection results, not physical robustness. Their tested tolerances do not automatically apply to a real robot. A still-attached object, changed identity or unsafe deviation aborts before the target call.

Hardware prerequisites

This guide does not authorize live actuation. First calibrate camera/robot transforms, pin real arm/gripper/tool geometry and limits, establish tracking/pose uncertainty, and validate controller contact behavior. Support-contact transport may require force/slip assessment absent from kinematic replay. A simulated rigid attachment cannot establish physical grasp stability.

Run offline kinematic checks, then a declared dynamics/controller stage where available, followed by separately authorized sparse hardware trials. Measure placement errors and choose a justified margin; the simulation's 5 mm is not a default hardware guarantee. Record observed target goal and retreat completion, not just commands.

What constitutes a result

Keep static-valid, kinematically verified, simulated-success and hardware-success fields distinct. A failed physical move cannot be undone by restoring a simulator snapshot and trying the next candidate while retaining a one-shot label. Hardware evidence supplements rather than replaces controlled explanation and order ablations.

15. Visual evidence, budget enforcement and reporting

Every gate needs scene-backed outputs. Save the clean input separately from overlays. Show all objects, target/goal, full robot, axes/units, proposed destinations, action order, evidence scope and actual state where applicable. Use written status labels as well as color.

Ghosts: draw a proposed pose as a transparent alternative linked to its source object. Never render all ghosts as simultaneously real obstacles. **Order:** show numbered proposed actions and the corresponding achieved-state timeline. **Explanation:** show the baseline and intervention, exact fixed fields, predicted effect and decided/unknown evidence. **Cost:** distinguish endpoint distance, predicted joint travel and measured joint travel.

Motion GIFs replay stored checked trajectories. Do not draw interpolated straight lines between endpoints as though a planner succeeded. For unknown cases display the recorded search stage and budget, not an invented collision or smooth path. Keep viewpoints and axis scales fixed in comparisons and inspect every exported figure for clipped links/fingers/text.

Budgets

Every collection plan declares node/evaluation caps, CPU budget, worker count, cancellation margin, wall-time safety guard and attempt ledger before outcomes. Monitor aggregate in-flight process-tree CPU, preserve termination records and identify wall-dependent results. Censoring produces UNKNOWN unless complete evidence already exists. Do not infer the exact cap-crossing root from aggregate totals without scheduling logs.

Stage report contents

Record document/protocol/schema versions; branch/source/profile/config/split hashes; changed modules and responsibilities; all attempted cases and construction failures; component evidence; unique witnesses and certificate-record counts; planner calls; actual CPU/time; tests and replay results; visual paths; input and target-feedback audits; scope changes; and the next gate's explicit readiness decision.

Use one new report number and one flat run directory per stage. Keep prior reports/archives unchanged. Save checkpoints with preprocessing, model and cost-policy definitions, label-mask version and all hashes needed to reject incompatible reloads. A successful file write or passing test suite is not a successful research gate.

Finishing a stage

Report outcomes plainly, including no improvements. Follow the established repository synchronization authorization, preserve unrelated work and never force-push. Stop at the gate boundary. Do not expand budgets, change root mixtures or start training just because the previous run was inconvenient.

16. Implementation checklist and exact changes from v1.7

Before model training

1. V2 proposals have one canonical ordered sequence; derived assignments cannot disagree with it. Empty and padded representations are unambiguous.
2. Verifier follows the proposed order; original V1 order search remains separate. Target callback is unreachable before completed repair.
3. Initial joints, robot geometry, gripper and constraints reach the task/cost scorer. Teacher intermediate joints do not leak into proposal inputs.
4. Ghosts substitute poses; omission restores stay; deletion is separately labelled. Counterfactual comparisons share the original appearance anchor.
5. Ordered, assignment-wide and trajectory-only evidence scopes are different. Unknown labels are masked correctly.
6. Scene and omission effect outputs are required, with baseline/intervention provenance and no automatic necessity claim.
7. Motion-cost labels use the declared first relocation-witness policy and normalized unwrapped joint travel, not endpoint distance.
8. Gibbs, swap and birth/death updates satisfy exact tiny-space checks, including the Hastings ratio and variable candidate counts.
9. G4D and G4C demonstrate actual decided order/effect/cost coverage. Failed quotas remain failed in their original run.
10. Training/ranking use the specified distinct feasibility and sampling energies; thresholds and cost settings are frozen on validation.

What changed

V1.7 had endpoint prediction, external order selection, optional causal diagnostics and no explicit robot-motion-cost head. V1.8 requires learned ordered proposals, anchored counterfactual and omission outputs, fixed-order execution checking, sequence-level labels, a recurrent ordered scorer, a joint-travel head, and mixed Gibbs/Metropolis search. It preserves the original no-target-feedback boundary and adds the reported safe-retreat completion contract.

The original G0–G10 progression remains, with G3.5 and G4A/B/C preserved and G4D added as the design/data migration checkpoint. Old fixed-route minima, V1 assignment labels and new V2 ordered labels remain separate. The guide now includes explicit readiness targets, failure conditions and claim-specific comparisons for the new outputs.

Definition of completion

The project is not complete merely because energy decreases. Completion of the research claim requires measured improvement on held-out infeasibility prediction, scoped intervention explanations and complete robot-executable correction, with costs and unknowns reported. If one output is not supported, state that limitation rather than removing it from the architecture description.

Companion: Methodology specification v1.6. This guide is an implementation contract. No new run, training outcome, certificate or hardware success is claimed by the revision itself.